

Coverage-Guided Search-Based Test Generation for a Tennis Scoring Engine: A Two-Regime Comparison Against Random Testing

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Abstract—Search-based test generation guided by code coverage is a well established technique, yet its advantage over plain random testing is known to depend on the difficulty of the search problem. We study a coverage-guided evolutionary test generator for a tennis scoring engine, a compact state-machine subject under test (SUT) with branching logic for deuce, tie-breaks and three tournament rulesets. We compare it against a random baseline and an ablation that keeps the evolutionary machinery but removes the coverage-guided fitness, under a two-regime budget design with $N=30$ repetitions, measuring branch coverage and mutation score with Mann-Whitney U tests and Vargha-Delaney A_{12} effect sizes. Random testing is competitive with, and frequently superior to, the guided search at every budget. The value of the coverage-guided fitness emerges instead against the unguided evolutionary baseline, where it grows with the budget and reaches a large effect on mutation score. We discuss why coverage saturation on small subjects neutralises guidance relative to random testing, and what this implies for when search-based testing is worth its cost.

Index Terms—search-based software testing, branch coverage, mutation testing, random testing, ablation study, empirical evaluation.

I. Introduction

Automated test generation aims to produce test suites that exercise a program thoroughly with little human effort. Search-based software testing (SBST) casts this as an optimisation problem in which candidate test inputs are evolved to maximise a coverage-oriented fitness function [1], [2]. A recurring and practically important question is whether the guidance provided by such a fitness function actually outperforms undirected random testing, which is known to be surprisingly competitive on simple domains [3].

We investigate this question on a tennis scoring engine: a deterministic state machine whose control flow encodes the non-trivial rules of tennis (game, deuce/advantage, set, tie-break) and three tournament rulesets that differ in the decisive-set tie-break. The SUT is small but branch-rich, which makes it a clean testbed for isolating the contribution of coverage guidance. We compare three generators at equal budget: a coverage-guided evolutionary algorithm, a random baseline, and an ablation that retains the evolutionary structure but replaces the fitness with a

constant, thereby isolating the effect of the fitness function itself.

We frame the study with the Goal-Question-Metric paradigm. The goal is to assess whether, and under which conditions, coverage-guided search improves test effectiveness over random testing for the tennis engine. We measure effectiveness through branch coverage (a structural proxy) and mutation score (a fault-detection proxy [4]). The research questions are:

- RQ1. Does coverage-guided generation achieve higher branch coverage and mutation score than random testing at equal budget?
- RQ2 (ablation). Is any observed contribution attributable to the coverage-guided fitness, or merely to the evolutionary structure?
- RQ3 (regime). How does the advantage of guidance vary with the search budget?

The main finding is a two-regime picture. Random testing is competitive or superior on both metrics across all budgets, because coverage saturates quickly on this small SUT. The fitness function nonetheless has a real and growing effect relative to the unguided baseline, becoming a large effect on mutation score at medium-to-high budgets. We report all numbers honestly, including the negative result for RQ1.

II. Related Work

Search-based test generation has been studied extensively since the survey of McMinn [1]. Tools such as EvoSuite [2] generate whole test suites for object-oriented programs by optimising a coverage criterion, and many-objective formulations such as MOSA [5] treat individual coverage targets as separate objectives to steer the search toward hard-to-reach branches. The competitiveness of random testing against more sophisticated strategies is a recurring theme: Arcuri et al. [3] give theoretical and empirical conditions under which random testing is effective, showing that on many problems the probability of covering targets by chance is high enough to make guidance unnecessary. Our study is methodologically aligned with the guidelines of Arcuri and Fraser [6] for assessing randomised algorithms, using repeated runs, non-parametric tests and standardised effect sizes [7]. We adopt mutation analysis

as a fault-detection proxy following the evidence of Just et al. [4] that mutant detection correlates with real fault detection. Our contribution is not a new algorithm but a controlled, ablated comparison on a compact, fully specified SUT, together with a two-regime budget analysis that isolates when coverage guidance pays off.

III. Subject Under Test

The SUT is a single-module tennis scoring engine (src/tennis_engine.py). The public API follows an internal contract:

- `TennisMatch(ruleset)` with `ruleset` $\in$ {wimbledon, usopen, ausopen};
- `play_point(winner)` drives the engine with `winner` $\in$ {"A", "B"}, raising `ValueError` for an invalid winner and `RuntimeError` if called after the match ends;
- `score()` returns a dictionary with exactly the keys `points`, `games`, `sets`, `in_tiebreak`, `is_over`, `winner`.

The engine implements best-of-three matches. A game is won at four points with a two-point margin (deuce/advantage); a set is won at six games with a two-game margin, at 7–5, or through a tie-break at 6–6. The three rulesets differ only in the decisive-set tie-break target: Wimbledon and Australian Open use a 10-point tie-break (two-point margin), the US Open a standard 7-point tie-break. Scoring logic is written with explicit if/elif chains, intentionally producing a diverse set of branches for deuce, advantage, the 7–5 and 7–6 set endings, and the per-ruleset tie-break targets. Under `coverage.py` in branch mode the module exposes 66 branch arcs in total. A test input is simply a sequence of point winners, i.e. a list over {A,B}; the search space is the set of such sequences up to a maximum length.

IV. Approach

A. Random baseline

The random generator emits budget independent sequences of point winners; each sequence has a uniformly random length in [4, 400] and each point is drawn uniformly from {A,B}. It is fully deterministic given the seed (a local `random.Random(seed)` instance is used throughout), which guarantees reproducibility.

B. Coverage-guided search

The search generator is an evolutionary algorithm over sequences of point winners. It initialises a population of random sequences, then repeatedly: (a) executes each candidate and (b) scores it; (c) selects elites by fitness; (d) produces offspring by mutation and crossover. The mutation operators are point flip, insert, delete, contiguous block-replace and suffix extend; single-point crossover recombines two elites. Every evaluated sequence is accumulated in an archive, which constitutes the returned test suite. The configuration is summarised in Table I.

TABLE I
Evolutionary algorithm configuration.

Parameter	Value
Population size	8
Elite ratio	0.3
Crossover probability	0.3
Sequence length range	[4, 400]
Budget (fitness evaluations)	{20, 60, 200}
Repetitions per cell	$N = 30$

The fitness of a candidate combines its own branch coverage with a novelty bonus for arcs not yet in the archive:

$$fitness(s) = |C(s)| + |C(s) \setminus A|,$$

where $C(s)$ is the set of branch arcs covered by s and A is the running archive of covered arcs. The own-coverage term is deliberate: a fitness based on novelty alone collapses to zero for every candidate as soon as the archive saturates, which on a small SUT happens within the first generation and removes all selection pressure. The combined form keeps a non-zero gradient throughout the search and is what makes the guided search differ from the ablation in practice.

C. Ablation

The ablation generator is the same algorithm as the search generator, invoked with a constant fitness ($\equiv 0$). It is implemented by delegating to the search generator and injecting the constant fitness function, so that population, selection, mutation and crossover operators, number of generations, and random-number usage are identical; only the fitness differs. This guarantees that any difference between search and ablation is attributable to the coverage-guided fitness and to nothing else. With a constant fitness the selection has no signal and the search degenerates into an unguided walk over the operator space. We verified empirically that the two generators produce distinct suites at budgets that allow at least one generation of offspring, and that all three generators are deterministic given the seed.

D. Oracle of the generated tests

Mutation score is only meaningful with a strong oracle. We therefore translate each generated sequence into a test that, after every `play_point` call, asserts state invariants of the contract rather than hard-coded expected scores (which are unknown a priori for generated sequences). The invariants include: the exact key set of `score()`; integer ranges for sets ($0 \leq \cdot \leq 2$, $\text{sum} \leq 3$) and games ($0 \leq \cdot \leq 7$); admissible point tokens ({0,15,30,40, Ad} in a game, non-negative integers in a tie-break); consistency between `is_over` and `winner`; monotonicity of the aggregate set and (within a set) game counts; terminal behaviour (a further point after the match is over must raise `RuntimeError`); and rejection of invalid input. A null oracle (e.g. asserting

only that `score()` is not `None`) can kill only mutants that crash the execution; it cannot detect mutants that silently corrupt the score, which is precisely what the invariant oracle catches. The choice of oracle is therefore a first-order factor for the mutation score and is held identical across all three generators.

E. Coverage measurement

Branch coverage is measured with `coverage.py` in branch mode, targeting only the engine module, through the JSON report API. Each measurement uses a fresh Coverage instance and reports from in-memory data, so there is no coverage leakage across runs. The total number of branch arcs (66) is constant across all runs, as expected for a fixed SUT.

V. Experimental Setup

We run a full factorial design over three generators (random, search, ablation), three rulesets, and three budgets, with $N=30$ independent seeds per cell, for 810 runs in total. The two-regime design contrasts a low budget, where random testing does not saturate coverage, against a high budget; we additionally include an intermediate budget to trace the transition. The budgets are $\{20, 60, 200\}$ candidate evaluations.

Budget is operationalised as the number of candidate sequences a generator may consider, and all generators return a suite of $\approx \text{budget}$ sequences, so the final measured suites have the same size. Tooling: `coverage.py` (branch mode), `mutmut v3` for mutation analysis, `scipy` for the statistics; Python 3.10. For each run we record branch coverage, mutation score, and wall-clock time.

a) Statistical analysis: For each (budget, ruleset, metric) we compare search against random and against ablation using the two-sided Mann–Whitney U test ($\alpha = 0.05$) and the Vargha–Delaney A_{12} effect size [6], [7]. We report A_{12} with the conventional thresholds negligible ($|A_{12} - 0.5| < 0.056$), small (< 0.14), medium (< 0.21) and large otherwise. $A_{12} > 0.5$ favours search; $A_{12} < 0.5$ favours the comparison baseline.

b) Threats to validity: Construct. Branch coverage is a structural proxy and mutation score a fault proxy; both are imperfect, though mutation score is an established predictor of real fault detection [4]. Branch coverage is moreover operationalised by driving the SUT through `play_point` only: the branches of `score()` (point/score formatting) and the error paths are not exercised by this measurement harness, so the reported figures underestimate the coverage actually achieved by the generated suites, which call `score()` after every point and reach ≈ 0.97 ; since all generators are measured identically, this does not affect their relative ranking. Finally, two arcs—the 7–5 set-win case in `_check_set_over`—are dead code, dominated by the general “ ≥ 6 games with a two-game margin” rule and hence logically unreachable, so they cap the attainable coverage below 1.0 for every generator.

TABLE II

Median branch coverage and mutation score per generator and budget (representative ruleset; rulesets agree to within rounding).

Metric	Budget	random	search	ablation
Branch cov.	20	0.727	0.682	0.682
	60	0.773	0.727	0.727
	200	0.773	0.773	0.773
Mutation	20	0.398	0.377	0.380
	60	0.435	0.427	0.399
	200	0.451	0.448	0.424

TABLE III

Statistical comparison (search vs. baseline). p : Mann–Whitney U (two-sided); A_{12} : Vargha–Delaney effect size (> 0.5 favours search). Representative ruleset.

Comparison	Budget	p	A_{12}	Effect
Branch coverage				
search vs random	20	0.0005	0.250	large (random)
search vs random	60	0.0002	0.258	large (random)
search vs random	200	0.042	0.433	small (random)
search vs ablation	20	0.93	0.507	negligible
search vs ablation	60	0.21	0.588	small (search)
search vs ablation	200	0.015	0.642	medium (search)
Mutation score				
search vs random	20	$5.4e-5$	0.197	large (random)
search vs random	60	0.070	0.364	small (random)
search vs random	200	0.0004	0.240	large (random)
search vs ablation	20	0.51	0.451	negligible
search vs ablation	60	$2.4e-5$	0.818	large (search)
search vs ablation	200	$< 10^{-6}$	0.882	large (search)

Internal. Budget is operationalised as the number of candidate sequences, so the measured suites have equal size; the coverage-guided search, however, executes the SUT once per fitness evaluation ($\approx \text{budget}$ additional SUT executions) to obtain the coverage signal, a cost not paid by the random baseline. This asymmetry, if anything, favours the baseline, because at equal budget the guided search returns fewer unique sequences (mutation and elitism create duplicates); the comparison is therefore fair to conservative with respect to search. External. A single, small SUT limits generalisability; results may differ on larger programs with harder-to-cover branches. Conclusion. We mitigate randomness with $N=30$ repetitions and non-parametric tests with effect sizes.

VI. Results

Coverage and mutation-score medians are essentially invariant across the three rulesets (the rulesets differ only in a rarely-reached decisive tie-break); we therefore report representative medians in Table II and the statistical comparisons in Table III, noting that all three rulesets agree on significance and effect-size class. Figures 1 and 2 show the budget curves, and Fig. 3 the distribution at the highest budget.



Fig. 1. Mutation score (median $\pm$ IQR) versus budget. The guided search tracks but does not exceed random; it separates clearly from the unguided ablation as the budget grows.

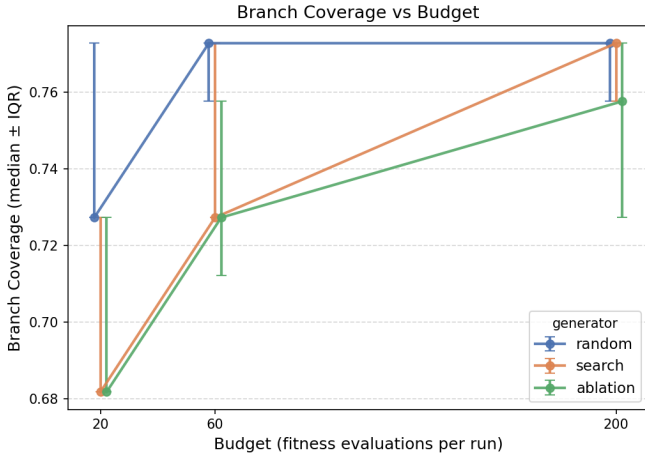


Fig. 2. Branch coverage (median $\pm$ IQR) versus budget. All generators converge as coverage saturates at high budget; random leads at low and medium budget.

A. RQ1: search vs. random

Coverage-guided search does not outperform random testing. On branch coverage, random is significantly better at budgets 20 and 60 (large effect, $A_{12} = 0.25$ and 0.26) and the two converge at budget 200 ($A_{12} = 0.43$, small). On mutation score random is competitive or superior at every budget ($A_{12} = 0.20$ – 0.36 favouring random); the gap is smallest and statistically non-significant at budget 60 ($p = 0.07$). The answer to RQ1 is therefore negative for this SUT.

B. RQ2: role of the fitness (ablation)

The coverage-guided fitness has a clear effect once the search runs long enough to breed offspring. On mutation score, search beats ablation with a large effect at budgets 60 and 200 ($A_{12} = 0.82$ and 0.88 , $p < 10^{-4}$), and on

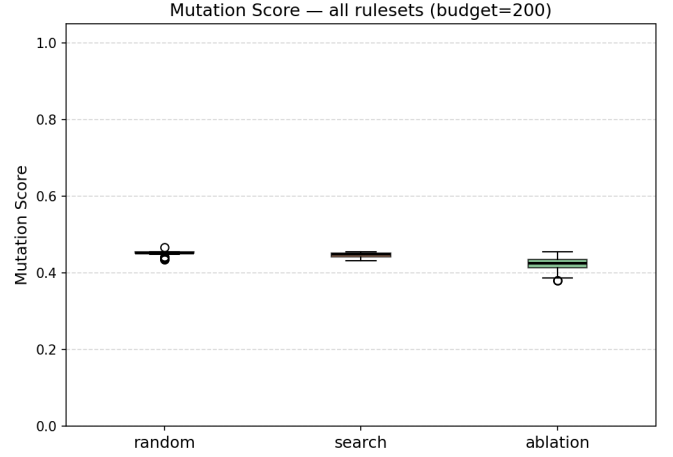


Fig. 3. Mutation score distribution at budget 200 (all rulesets). The guided search and random overlap, while the unguided ablation is clearly lower.

branch coverage with a medium effect at budget 200 ($A_{12} = 0.64$, $p = 0.015$). At budget 20 the two are statistically indistinguishable ($A_{12} \approx 0.5$), because the budget is exhausted before the first generation of guided offspring is produced. The fitness function thus provides a genuine, increasing contribution—just not enough to overtake random.

C. RQ3: effect of budget

The advantage of guidance over the unguided baseline grows monotonically with budget: negligible at 20, small-to-large at 60, large at 200 (Table III, Fig. 1). Against random, the picture is the opposite of the naive expectation: increasing the budget lets random saturate coverage and closes any gap rather than opening one.

D. Cost

Guidance is not free. Averaged over runs, the search generator takes about +18%, +22% and +28% wall-clock time over random at budgets 20, 60 and 200 respectively (e.g. 72.9s vs. 57.1s at budget 200), reflecting the extra SUT executions performed to compute the fitness.

VII. Discussion

a) Why random saturates and neutralises guidance: The SUT is small and its branches are reachable by moderately long random point sequences: a single random sequence frequently plays a full match and exercises deuce, set and tie-break logic. With independent sampling, a handful of such sequences already cover most of the 66 arcs, so coverage saturates around 0.77 regardless of guidance. This is the classic regime in which random testing is hard to beat [3]: when the probability of covering a target by chance is high, a fitness gradient adds little. The guided search, by concentrating on a converging population, even explores fewer distinct states at equal budget, which explains why it does not exceed random.

b) Why coverage and mutation score can diverge: Covering the branch that a mutant lives on is necessary but not sufficient to kill it: the mutated state must also propagate to an assertion. The search’s tendency to converge reduces the diversity of game states reaching the oracle, so even when branch coverage is matched, the guided suite may kill slightly fewer mutants than a more diverse random suite. This decoupling between a structural proxy and a fault proxy is well documented [4] and is visible here as the small but persistent random advantage on mutation score (Fig. 3).

c) When guidance pays: The ablation isolates what the fitness actually buys: a large mutation-score gain over an unguided evolutionary search at sufficient budget. Guidance therefore does steer the search toward more effective tests; its failure to beat random is a property of the subject (cheap-to-cover), not of the fitness. We would expect the ranking to invert on subjects where targets are rare relative to the budget, i.e. where random rarely hits the hard branches and a gradient becomes decisive [1], [5].

d) Practical implication: Search-based testing is worthwhile when the search space is large or hard relative to the available budget; on cheap-to-cover subjects it adds a measurable time overhead ($\sim 28\%$ here) without improving effectiveness over random. Practitioners should size the budget and choose between random and guided search according to how easily targets are reached.

VIII. Conclusion

We presented a two-regime empirical comparison of a coverage-guided evolutionary test generator, a random baseline, and an unguided ablation on a tennis scoring engine. The contribution is the two-regime finding itself: random testing is competitive or superior on this small SUT at all budgets, while the coverage-guided fitness yields a real, budget-increasing improvement over the unguided baseline—large on mutation score—without overtaking random. The result is a concrete instance of the random-testing-competitiveness phenomenon, with an ablation that pinpoints where the fitness contributes. Limitations include a single small SUT and a mutation-agnostic fitness. Future work includes a mutation-aware fitness, a larger and harder SUT with rarely reached branches, and a denser sweep of intermediate budgets to locate the crossover, if any, between guided search and random testing.

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